

New Jersey



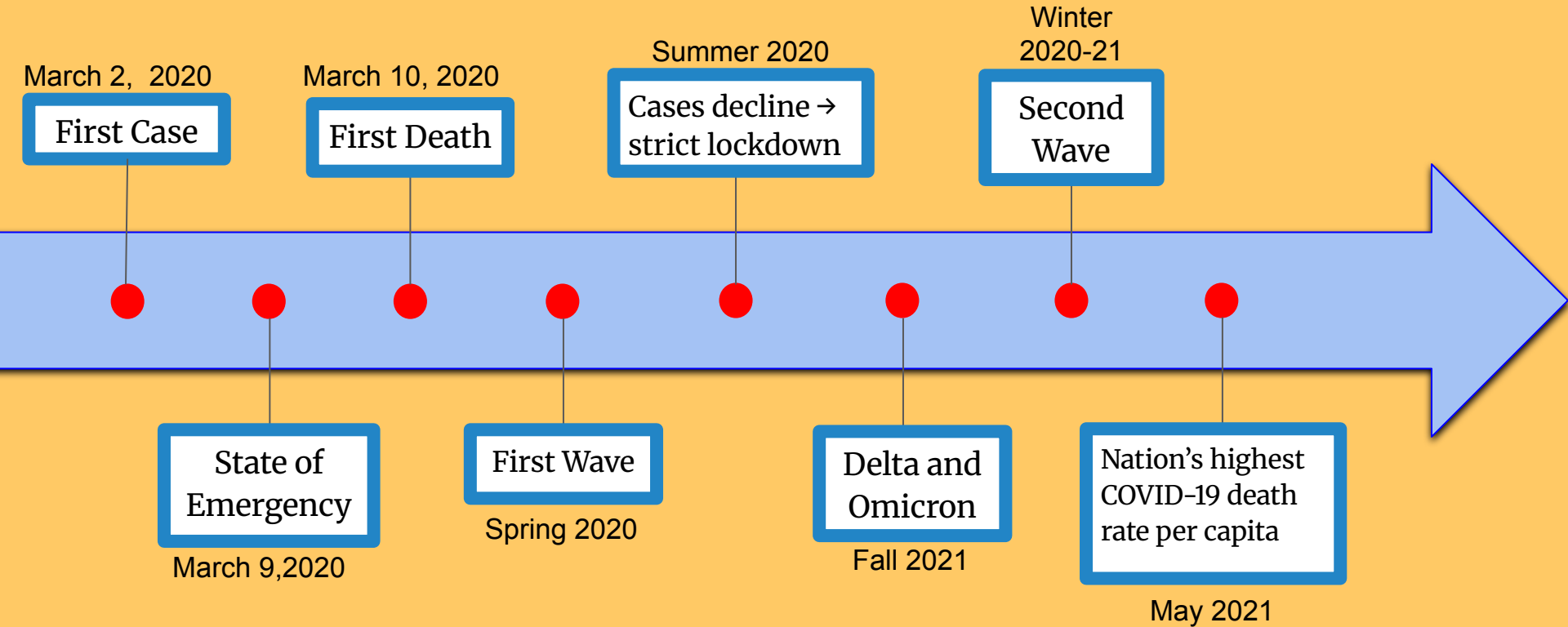
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Background On New Jersey

- 9.501 million
- Most densely populated US state
 - (~1,253 people/mi²)
- 21 Counties
- Largest county: Burlington
- Smallest County: Hudson
- Urban counties = smaller; rural = larger
- Capital: Trenton
 - Mercer County
- Close to New York
- “Diner Capital of the World” (450)
- 114.6 million tourists in 2022
 - 18 million more than in 2021
- Mask Mandate → 53% disliked
- Vaccine → 30% not getting
- 29% wanted normal life



TIMELINE



Data and Analysis Overview

Research Question:

- What county-level factors explain the speed of early COVID-19 spread?
- How do mobility, urbanization and demographics affect outbreak patterns?

Data Sources:

- **COVID-19 case & death data** from the New York Times GitHub
- **Demographic & socioeconomic variables** from U.S. Census (ACS tables DP05, B08134, B28002)
- **Hospital bed counts** from the American Hospital Directory
- **Uninsured rates** from the Census SAHIE tool
- **Population density, commute times, income** from Census and NJ sources
- **Merged by county** using `left_join()` in R to unify dataset
- Final dataset included **21 counties** with complete observations

Explanatory Variables

Variable	Description	Mean
Median Household Income	Represents the median income of individuals within each county	84634.5
Population Density	Measures how densely populated a county is	928.5
Unemployment Rate	Percentage of the labor force that is unemployed	7.5
Percent of Commuters	Percent of workers who commute 60+ minutes	13.2
Airport Access	Binary indicator if county contains a major airport	
Percent of Internet Access	Percent of households with Internet Access	91.8
Proximity to Major Airports	Categorizes counties based on how close they are to commercial airports	
Percent of People Age 65+	High Risk Population	23.7
Percent Uninsured	The percentage of the county's population under age 65 that does not have health insurance coverage	17.7
Percent White	Percent of White People	51.94
Percent Black	Percent of Black People	12.2
Percent Hispanic	Percent of Hispanic People	21.9
Percent Asian	Percent of Asian People	9.8
Beds per Capita	Total Staffed Beds per 1,000	2.12

Outcome Variables

Variable	Description
Total Cases	Number of total cases
Total Deaths	Number of total deaths
Cases per 100k	Number of cases per 100k population
Deaths per 100k	Number of deaths per 100k population
Days to 100th Case	How many days it took from the first reported case to the 100th case

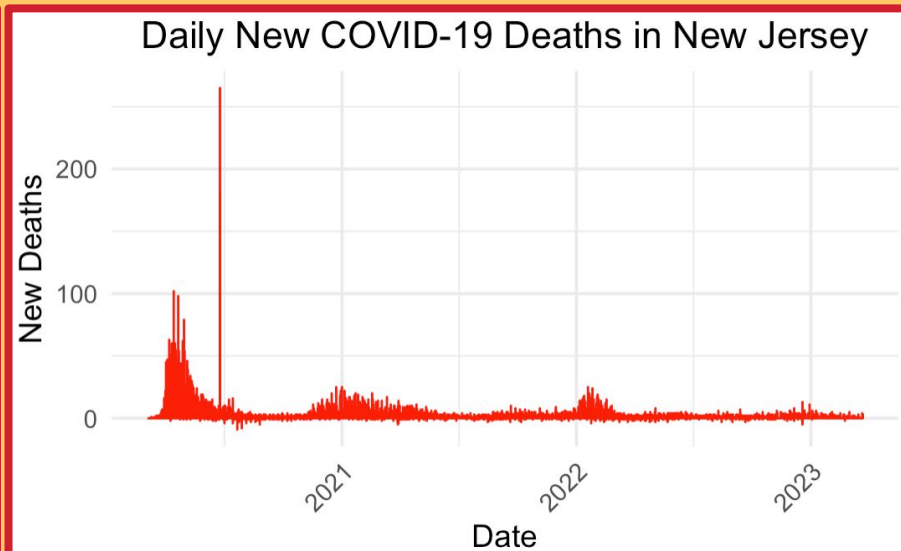
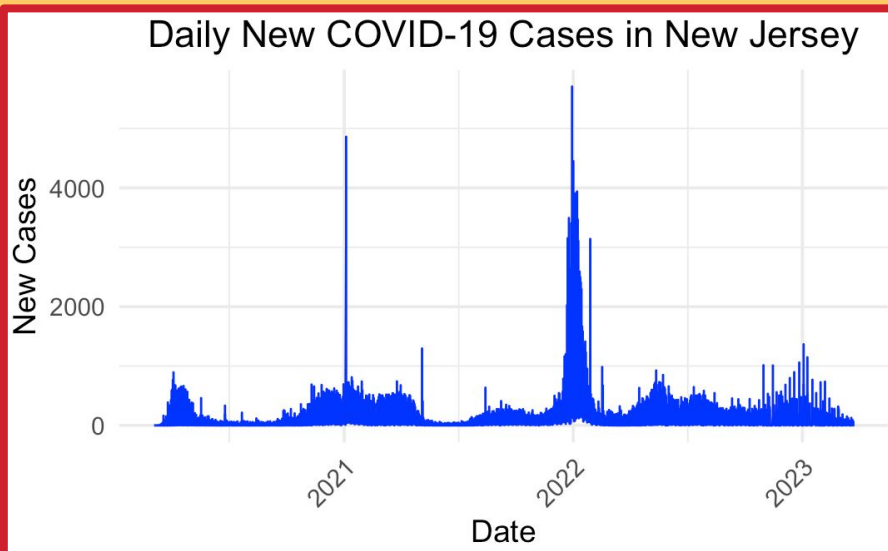
Summary Statistics

Variables	Mean	SD	p0	p25	p50	p75	p100
Final Cases	145514	94,847	18,650	52,987	142,308	216,686	327,464
Final Deaths	1719	1173	265	665	1,379	2,500	3781
Percent of Long Commuters	13.3	4.14	7.37	9.48	13.4	16.7	20.8
Median Income	84,635	25,050	45,000	72,000	80,000	90,000	133,534
Population Density	929	452	300	500	950	1200	2000
Unemployment Rate	7.6	6.38	3.9	5.1	6	7.3	34.4
Percent with Internet	91.9	2.37	87.8	90.7	91.7	94.0	95.5
Percent White	58.1	18.7	27.9	41.9	54.5	75.2	84.4
Percent Black	11.0	8.10	2.27	5.29	9.96	15.8	36.0
Percent Hispanic	19.4	10.9	0.654	10.7	17.0	23.0	43.4
Percent Asian	7.50	6.68	7.84	2.96	5.51	10.6	25.0
Percent 65 and over	17.7	3.36	12.4	15.7	17.8	18.9	28.6
Days to 100	14.6	8	8	11	14	18	26
Cases per 100k	32,128	2,632	26,880	30,188	32,443	34,244	37,429
Deaths per 100k	376	63.1	239	345	372	408	509
Percent Uninsured	23.7	7.76	14.1	16.8	21.1	27.4	38.9
Total Staffed Beds per 1,000	2.12	1.14	0.95	1.45	1.83	2.51	6.14

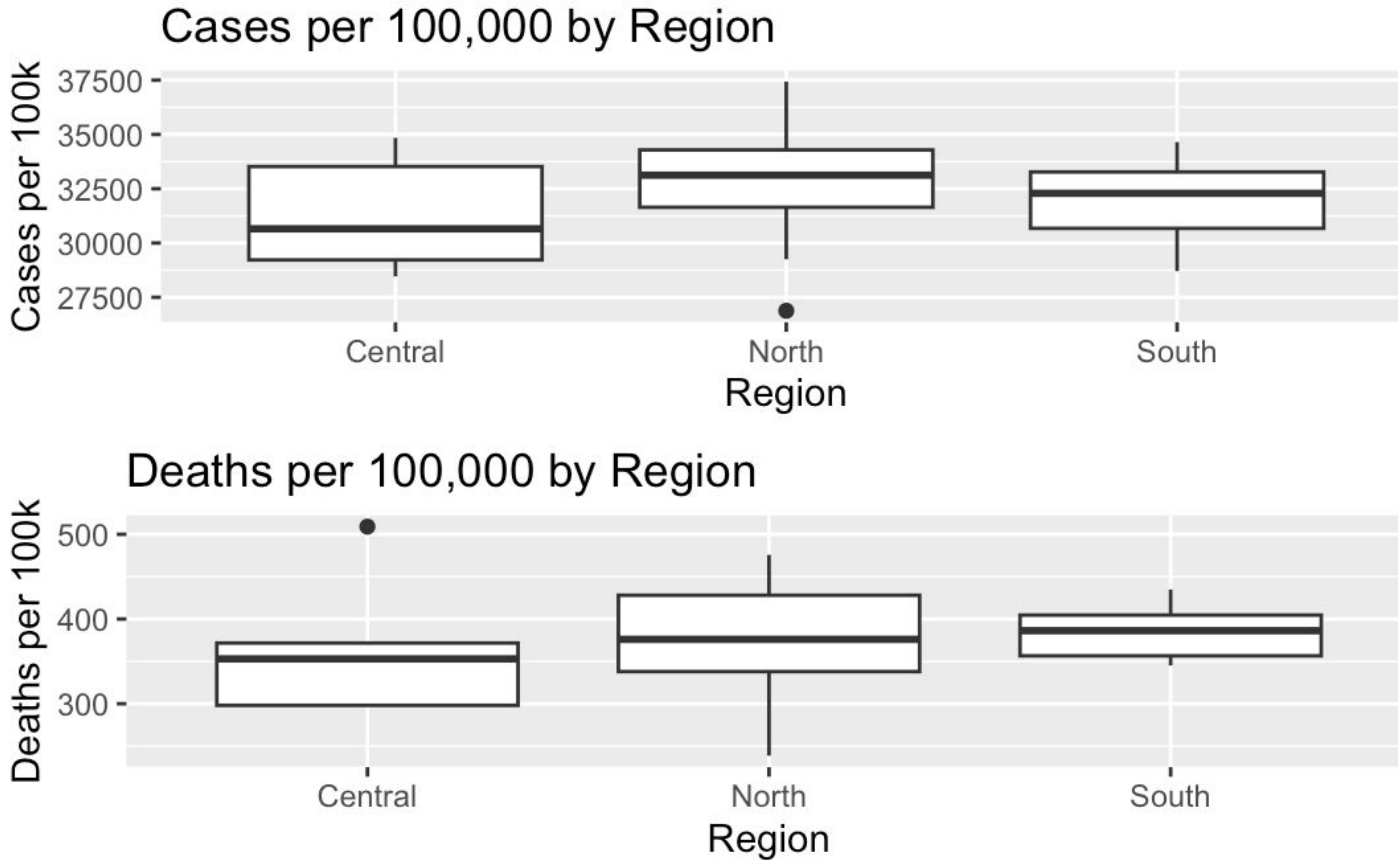
Descriptive Analysis

COVID-19 Trends Over Time

- 2022, Omicron variant prevalent
- For deaths, giant spike happened on 06/25/2020 in Bergen County of 265 new deaths and Essex county following with 247 that same day



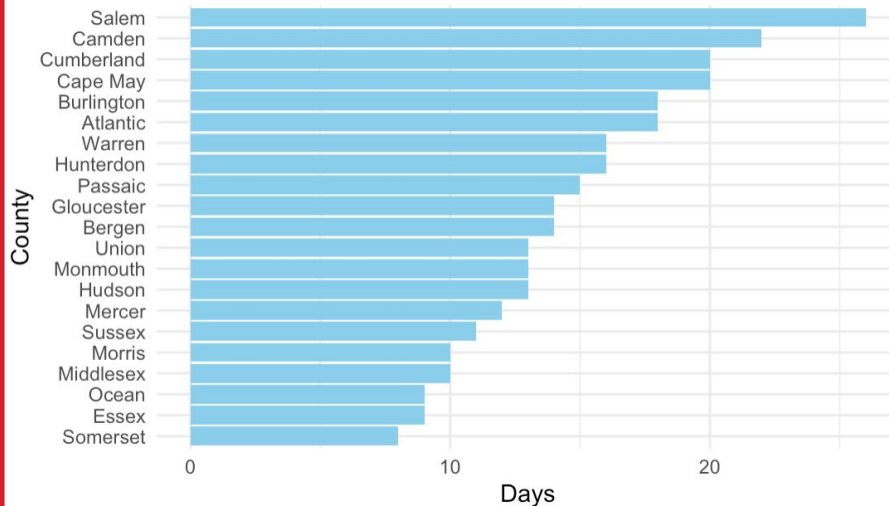
COVID-19 Trends Over Time Continued



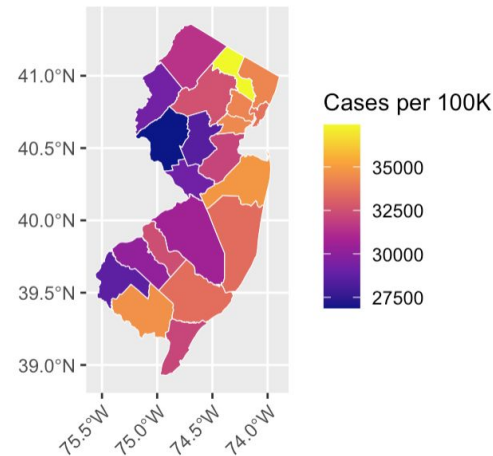
Geographic Impact by County

- Bergen, Essex, Passaic hit hardest
 - Highest cases and deaths per capita
- Rural counties: fewer cases per capita in early stages (Salem)
- Late 2021 → gap narrowed and pandemic spread

Days to Reach 100 COVID-19 Cases by County

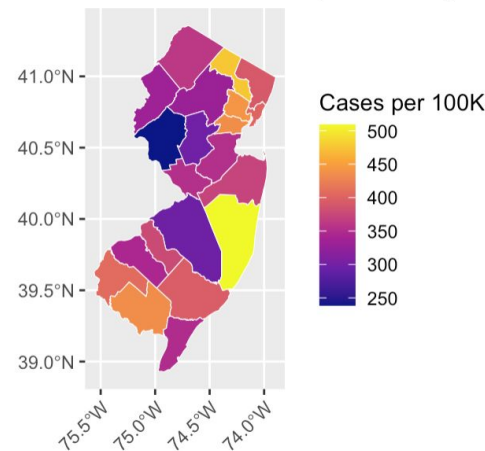


COVID-19 Cases per 100k by County



Source: Jersey COVID-19 and ACS Population Data

COVID-19 Deaths per 100k by County



Source: Jersey COVID-19 and ACS Population Data

Inferences

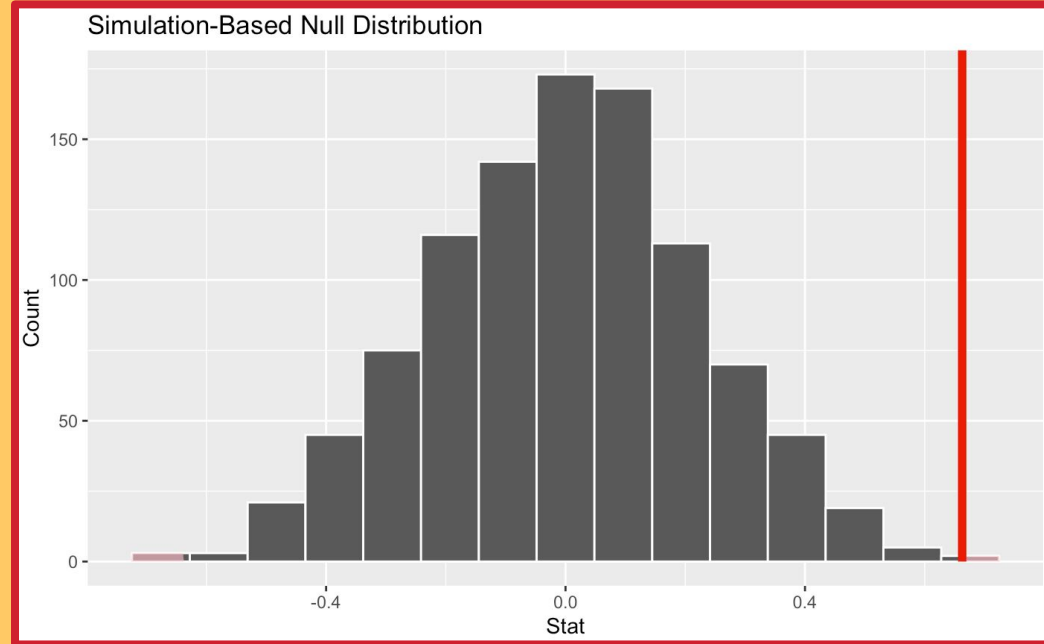
Commuter % vs. Days to 100 COVID-19 Cases

- Permutation test using the **slope of a linear model** as the test statistic
- **Null Hypothesis (H_0):** There is no association (slope = 0).
- **Alternative Hypothesis (H_1):** There is an association (slope $\neq 0$).
- The observed slope was -0.695
- p-value = 0.0034



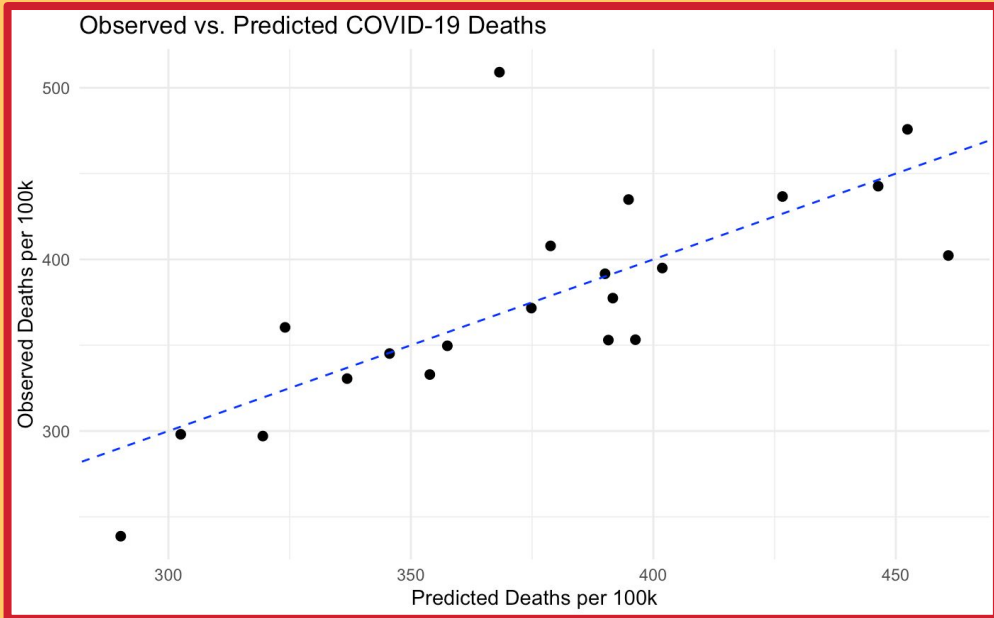
Correlation Between % Uninsured and Deaths per 100k

- Permutation test using the **Pearson correlation** coefficient as the test statistic
- **Null Hypothesis (H_0)**: There is no association between percent uninsured and deaths per 100k.
- **Alternative Hypothesis (H_1)**: There is an association (positive or negative).
- Counties with higher uninsured rates tended to experience higher COVID-19 death rates per capita (p-value = 0.004)



Multivariate Regression: Predicting deaths per capita

Outcome	Predictors	Estimate	Probability
Deaths per 100k	Intercept	-14.77	0.905
	Percent Uninsured	4.585	0.002
	Log Population Density	42.118	0.038



Conclusion & Discussion

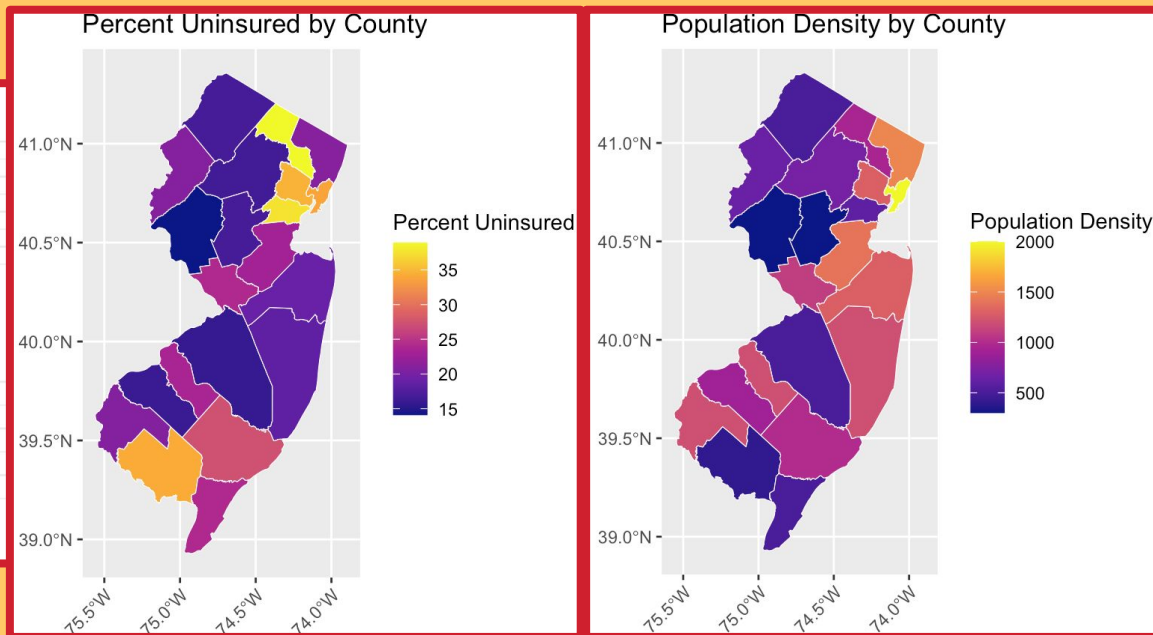
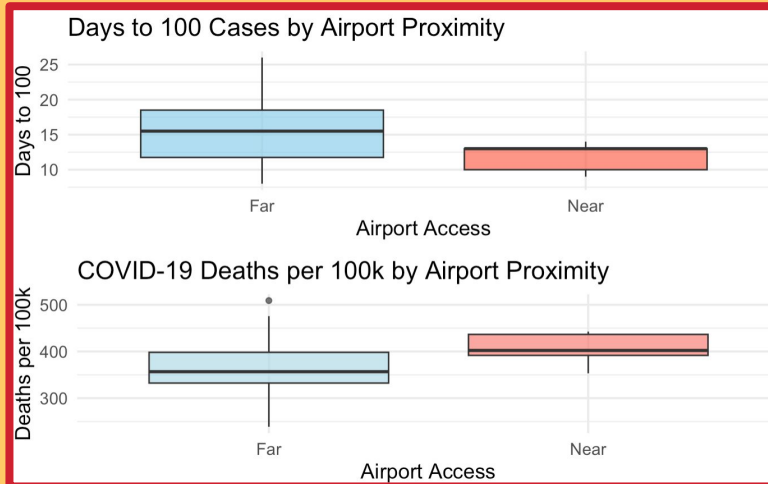
Key Findings

- Higher-density counties experienced higher infection rates
 - Density was not sole factor in increased transmission
- Higher uninsured rates were significantly associated with more COVID-19 deaths ($p = 0.004$)
- Strong evidence that commuting patterns were associated with the early speed of COVID-19 spread across counties ($p = 0.0034$)

COVID Outcomes vs. County Characteristics

- **Population Density:** Higher-density counties = higher infection rates, urban counties were hit hardest
- **Age Distribution:** Counties with older populations = higher COVID-19 mortality rates
- **Socioeconomic Factors:** Little correlation between median household income and overall case rates
- **Region:** Central region reached 100 COVID cases first, lowest beds per capita
- **Airport Access:** Near-airport counties hit 100 cases faster and had higher median deaths per 100k

Additional COVID Outcomes



Discussion

- Access to healthcare and living environments played critical roles in pandemic outcomes at the county level
- Other variables considered: hospital bed capacity, percent elderly, socioeconomic factors and racial/ethnic composition did not show significant associations in this dataset
- Data is observational, no causal conclusions
- Small sample size (21 counties) limits power
- Deaths and cases may be underreported in early stages

References

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